DATA ITEM DESCRIPTION

Title: Digital Twin Description (DTD)

Number: DI-SESS-82483
AMSC Number: F10554
DTIC Applicable: No
Preparing Activity: 19 (SSC)
Applicable Forms: N/A

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Use/Relationship:
The Digital Twin Description (DTD) describes the design and context of a Digital Twin (DT) to be delivered. The Data Item Description (DID) contains content requirements that would support the acquisition, production, inspection, engineering, logistics, maintenance, integration(s), and upgrade(s) of the DT to be delivered. The DID also contains types of requirements that shall be imposed on a DT to meet most work tasks that would be described within a contract or solicitation.

A DT is a computerized representation (integrated set of models) that serves as a real-time[1] digital counterpart of a physical object or process[2, 3].

A DT is also a virtual replica of a physical entity that is synchronized over time. A DT exists to replicate configuration, performance, or history of a system[4].

In addition to the definitions, how a DT is federated with other DTs across the lifecycle impacts what constitutes "a delivered DT". Thus, this DID is written to consider a DT in any of the following three ways:

1) How the delivered DT interacts with other DTs that support the same lifecycle functions and the same system of interest (e.g., DT at "Facility A" vs DT at "Facility B").
2) How the delivered DT interacts with other DTs that support different lifecycle functions for the same system of interest (e.g., DT for "System A Design" vs. DT for "System A Operations").
3) How the delivered DT interacts with other DTs that support the same lifecycle functions for different systems of interest (e.g., DT for "System A manufactured at Time A" vs DT for "System A manufactured at Time B").

The following are additional points of consideration for this DID.

a) The DTD is used as the basis for implementing and communicating various aspects of the DT. It provides the acquirer visibility into the life cycle of the DT and its context and provides information needed for DT support.

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[1] "Real-Time" to be taken as relative to the need and use cases.
[2] Definition taken from DoD Instruction 5000.97: Digital Engineering – dated December 21, 2023.
[3] In the context of this DID, a Digital Shadow (unidirectional interaction) with a human-in-the-loop to facilitate digital-to-physical twin interactions shall be considered a Digital Twin (bidirectional interaction).
[4] Definition taken from MIL-HDBK-539: Digital Engineering and Modeling Practices – dated December 6, 2022.

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b) If the DTD is delivered as a description model (i.e., system architecture or engineering model), this DID may be supplemented with a Model Based Systems Engineering Development Plan (MDP) (DI-SESS-82380).

c) If the DTD is delivered as a description model, the necessary fidelity of the description model shall be derived from the government-provided Use Cases of the DT and tailored into the DID in one of the following ways: 1) provided by the government, or 2) proposed by the contractor and agreed to by the government.

d) This DID is used when the contractor is tasked to define and record the design and context of a DT that supports a particular (or set of) System of Interest (SOI) lifecycle phase(s).

e) This DID may be supplemented with a Cybersecurity Supply Chain Risk Management (C-SCRM) Software Bill of Materials (SBOM) Requirements (DI-SESS-82433) and C-SCRM Hardware Bill of Materials (HBOM) Requirements (DI-SESS-82434) in support of Section 3.1.10.5.

f) This DID contains requirements spanning a number of DT use cases. Standard use of a DID allows for removal of statements to tailor for specific program needs.

Requirements:

1. **Reference Documents.** The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. **Format.**

   2.1. Description Format. The term “description” in this DID means a collection of data, regardless of its medium and is synonymous with DTD. The term “model” in this DID means a medium that follows a formal language. Therefore, the delivered description shall be a description model so that data required by this DID can be made individually accessible to stakeholders and their tools using automated techniques (such as Application Programming Interfaces (APIs)). The necessary fidelity of the description model shall be derived from the government-provided Use Cases of the DT and tailored into the DID in one of the following ways: 1) provided by the government, or 2) proposed by the contractor and agreed to by the government. In some cases, multiple DTs could be delivered, so the description(s) shall be organized such that the SOI and the lifecycle functions the DT will support are evident. If requirements are tailored out of this DID, the resulting description shall contain a corresponding placeholder appended with “This content has been tailored out.”

   2.2. Additional Description Format. The term “document” in this DID means a medium that follows a natural language. A description document in the contractor’s preferred format shall be an acceptable appendix so that data required by this DID can be made accessible to stakeholders when tools using automated techniques are not available.

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2.3. Implementation Format. The DT shall be delivered as the software implementation and hardware infrastructure (as needed) as well as the training, instruction, and expertise to deploy, duplicate, instantiate, and independently evaluate the implementation in government-directed facilities.

3. Content. The deliverable shall include content to address both description and implementation needs. If portions of the required content have already been provided by the government, the DTD shall indicate what information has been provided and show any traceability where government-provided information differs from what the contractor is using internally.

Note: For brevity, the term "Major System Component" is shortened to "component" and refers to Hardware Configuration Items (HWCIs) and Computer Software Configuration Items (CSCIs) relative to the term's modifier (e.g., Digital Twin Component is synonymous with HWCIs and CSCIs of the Digital Twin).

3.1. Description Content. The description shall include all data items that can allow an independent party to verify the definition, development, operation, and decommission of a delivered DT on a continuing basis. Sections 3.1.1 through 3.1.8 require descriptions about the DT system. Sections 3.1.9 through 3.1.16 require descriptions for its lifecycle context. While a description model is the preferred format, a narrative document may be appended to more strongly focus the narrative content and allow comprehension of the DT when a compatible software application is not available.

3.1.1. Scope. The scope shall include the following:

3.1.1.1. Identification. A full identification of the system and system components to which this DTD applies, including, as applicable, identification number(s), title(s), abbreviation(s), version number(s), and release number(s).

3.1.1.2. Digital Twin Overview. A summarization of the purpose of the DT to which this DTD applies. It describes the general nature of the DT and its key use cases; summarizes the history of DT development, deployment, operation, maintenance, and decommissioning; identifies the project sponsor, acquirer, user, developer, and support agencies; and identifies current and planned operating sites.

3.1.1.3. Description Model Overview. A summarization of the purpose and contents of the model-based description, including the security, privacy, or proprietary considerations associated with its use.

3.1.1.4. Description Document Overview. A summarization of the purpose and contents of the document-based description, including the security, privacy, or proprietary considerations associated with its use.

3.1.1.5. Description Model Fidelity. A definition of the degree to which the DTD represents the DT at DT defined levels of abstraction.

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3.1.2. Digital Twin Architecture Design Decisions and Principles. The description shall include the following to describe the DT architectural design.

3.1.2.1. Design Decisions. Decisions that govern the DT Architecture.

3.1.3. Digital Twin Components. The description shall include the following. If part (or all) of the design depends upon the maturity, state, or condition of the represented system, this dependency shall be indicated. Design conventions needed to understand the design shall be presented or referenced.

3.1.3.1. Components of the DT. CSCIs, data and their repositories, HWCIs, and operator decision points and each assigned a project-unique identifier.

3.1.3.2. Relationships of the DT Components. Identification, including a diagram of the static (such as “consists of”) relationship(s) of the DT components. Multiple relationships may be presented, depending on the design methodology.

3.1.3.3. Purpose of Each DT Component. Identification of the DT requirements allocated to each DT component.

3.1.3.4. Development Type of Each DT Component. Identification of each DT component’s development type (such as new, existing used as-is, existing to be reengineered, etc.) and status (such as being designed, testing, fit-for-use, accreditation, etc.). All development types and statuses are identified and defined with a unique identifier.

3.1.3.5. Modularity of Each DT Component. Identification of the following characteristics, as defined within the Modular Open Systems Approach (MOSA), for each DT component:

3.1.3.5.1. Interoperability. The ability to provide data, information, functions, or services to other DT components.

3.1.3.5.2. Maintainability. The ability to be retained in, or restored to, a specified condition by appropriately skilled personnel.

3.1.3.5.3. Extensibility. The ability to expand its capability through a well-defined process.

3.1.3.5.4. Composability. The ability to maintain minimal dependencies with DT components and leverage different DT components for new functionality.

3.1.3.5.5. Reusability. The ability to maintain certification, reliability, and performance in different assemblies of DT components.

3.1.3.6. Certifications Needed. Identification of the certification(s) that were needed to accredit each HWCI, CSCI, and data repository and element, as applicable.

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3.1.3.7. Licenses Needed. Identification of the license(s) and IP rights that will be needed (at an ongoing basis) to retain access to and use of each HWCI, CSCI, and data repository and element, as applicable.

3.1.3.8. System or Aggregate HWCI. For each system or other aggregate of HWCI identified for use in the DT (e.g., computer system, simulators, emulators, etc.), include a description of its resources (such as power, processor capability, memory, etc.). Each description, as applicable, identifies the configuration items that will use the resource, the allocation per system, conditions under which utilization will be measured, and the characteristics of the resource (manufacturer name and model number, type of resource, growth capabilities, etc.).

3.1.4. Digital Twin Behavior. The description shall include descriptions of the concept of operations, behavior, states, and processes through DT Components, including diagrams showing the dynamic relationship of the DT components. That is, how they will interact during DT assembly, storage, deployment, and operations.

3.1.4.1. Workflows of the DT. The specific sequence of steps, tasks, or actions to accomplish a goal or use case of the DT (including verification and testing workflows) and each unique sequence assigned a project-unique identifier.

3.1.4.2. States or Modes of the DT. The states or modes are identified as the specific sets of inputs the DT responds to, the sets of outputs it produces, and the triggers the DT can transition states or modes with. Each unique set assigned a project-unique identifier.

3.1.4.3. Purpose of Each Workflow, State, or Mode. Identification of the DT Use Cases allocated to it and whether it supports the physical-to-digital interaction (or vice versa).

3.1.4.4. States or Modes of the DT Components for each DT State or Mode. For each state or mode of the DT, include an identification of the state or mode of each DT component (identified in section 3.1.3.1: Digital Twin Components) and what workflows are involved during that state or mode.

3.1.4.5. Flow Through DT Components for each DT Workflow. For each workflow of the DT, include a description and diagram of the flow sequence through the DT components (identified in section 3.1.3.1: Components of the DT), what states or modes are involved, and what input data initiates the workflow.

3.1.5. Digital Twin Interface Design. The description shall include the following to describe and characterize the interfaces of the DT. Including both the interfaces among DT components and their interfaces with external entities such as other systems, other system DT, configuration items, and users.

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3.1.5.1. Interface Criteria. Identification of the criteria used to separate the DT into its components, the DT from external entities, and how the criteria impacted interface design decisions.

3.1.5.2. Interface Data Definition. Identification of the individual data elements and data element assemblies (records, files, reports, etc.) that interfacing entity(ies) will provide, store, send, access, etc. and their characteristics as follows:

3.1.5.2.1. Unique Identifier

3.1.5.2.2. Non-technical (natural language) name

3.1.5.2.3. Technical Name (e.g., field or data structure name in code or database)

3.1.5.2.4. Abbreviation or synonymous name(s)

3.1.5.2.5. Characteristics unique to Data Elements.

3.1.5.2.5.1. Data Type (alphanumeric, integer, etc.)

3.1.5.2.5.2. Size and format (such as length and punctuation of a char string)

3.1.5.2.5.3. Unit of measurement (such as meters, dollars, nano seconds)

3.1.5.2.5.4. Range or enumeration of possible values (such as 0-99)

3.1.5.2.5.5. Accuracy (how correct), precision (number of significant digits)

3.1.5.2.6. Characteristics unique to Data Assemblies.

3.1.5.2.6.1. Data Elements in the assembly and their structure (order, etc.)

3.1.5.2.6.2. Medium (such as disk) and structure of data elements on the medium

3.1.5.2.6.3. Visual and auditory characteristics of displays and other outputs (such as colors, layouts, fonts, icons, etc.)

3.1.5.2.6.4. Relationships among assemblies, such as sorting/access characteristics

3.1.5.2.7. Minimum maturity of data element or assembly (i.e., application to the DT's life cycle phase)

3.1.5.2.8. Priority, timing, frequency, volume, sequencing, and other constraints, such as whether the data element or assembly may be updated and whether business rules apply

3.1.5.2.9. Sources (setting/sending entities) and recipients (using/receiving entities)

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3.1.5.3. Interface Communication Method Definition. Identification of the communication methods that interfacing entity(ies) use for interfaces.

3.1.5.3.1. Unique identifier(s)

3.1.5.3.2. Communication links, bands, frequencies, media and their characteristics

3.1.5.3.3. Message formatting

3.1.5.3.4. Flow control (such as sequence numbering and buffer allocation)

3.1.5.3.5. Data transfer rate, whether it is periodic or aperiodic, and interval between transfers

3.1.5.3.6. Routing, addressing, and naming conventions

3.1.5.3.7. Transmission services, including priority and grade

3.1.5.3.8. Safety, security, and privacy considerations such as encryption

3.1.5.4. Interface Protocol Definition. Identification of the protocols that the interfacing entity(ies) use for the interface.

3.1.5.4.1. Unique identifier(s)

3.1.5.4.2. Priority and layer of the protocol

3.1.5.4.3. Packeting, fragmentation and reassembly, routing, and addressing

3.1.5.4.4. Legality checks, error control, and recovery procedures

3.1.5.4.5. Synchronization, including: connection establishment, maintenance, termination status, identification, and any other reporting features

3.1.5.5. Interface Identification and Diagrams. Identification of the interfacing entities by name, number, version, and documentation references, as applicable, and each assigned a project-unique identifier. The identification includes which entities have fixed interface characteristics (and therefore impose interface requirements on interfacing entities) and which are being developed or modified (thus having interface requirements imposed on them). If needed, one or more interface diagrams shall be provided, as appropriate, to depict the interfaces.

3.1.5.6. (Unique Interface Identifier). For each unique interface, include identification of the interfacing entities and descriptions for the interface characteristics of one or both of the interfacing entities.

3.1.5.6.1. Priority assigned to the interface by the interfacing entity(ies)

3.1.5.6.2. Purpose of interface to be implemented

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3.1.5.6.3. List of individual data elements involved at interface

3.1.5.6.4. List of data element assemblies involved at interface.

3.1.5.6.5. List of communication methods involved at interface.

3.1.5.6.6. List of protocols involved at interface.

3.1.5.6.7. Other characteristics, such as physical compatibility of the interfacing entity(ies).

3.1.6. User Stories, Use Cases, and Digital Threads. The description shall include the following:

3.1.6.1. List that includes a unique identifier and description of each derived assumption, clarification, or extension from the government provided:

3.1.6.1.1. Users (entities that interact with the DT)

3.1.6.1.2. User Stories (desired interactions with the DT by one or more User)

3.1.6.1.3. Use Cases (specific interaction implementations between the DT and Users).

3.1.6.2. Derived User Stories and Use Cases of the DT. Identification of the DT requirements and design decisions influenced by the derived Use Cases, and each assigned a project-unique identifier.

3.1.6.3. Static Relationships of the Derived Use Cases. Identification, including a diagram of the static relationship(s) of the derived Use Cases.

3.1.6.4. Purpose of Each Derived User Story and Use Case. Identification of the DT requirements allocated to each Derived User Story and Use Case, the objectives, and intended users.

3.1.6.5. Digital Threads[5, 6] for each Derived Use Case. For each derived Use Case, include identification of the layers of separation between the DT and the SOI and each sequence of layers with a unique identifier. This can be understood as the communication or analytic framework that any piece of information or data must be transformed through to produce actionable information. The accuracy of the data retrieved through the derived use case shall also be documented.

[5] Definition taken from DoDI 5000.97: An extensible and configurable analytical framework that seamlessly expedites the controlled interplay of technical data, software, information, and knowledge in the digital engineering ecosystem, based on the established requirements, architectures, formats, and rules for building digital models. It is used to inform decision makers throughout a system's life cycle by providing the capability to access, integrate, and transform data into actionable information.

[6] Definition taken from MIL-HDBK-539: A communication framework that connects a system's data across functional perspectives.

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3.1.7. Digital Twin Requirements and Requirements Traceability. The description shall include the following:

3.1.7.1. Stakeholder Needs and Requirements. Identification and descriptions of the derived assumptions, clarifications, or extensions from the government provided Requirements or Source Documents.

3.1.7.2. Requirements of the DT. Identification of all DT requirements and the unique requirements needed to integrate components of the digital twin, to federate unique digital twins, and to integrate with other pre-existing DT components to perform as a stand-alone DT and each assigned a project-unique identifier.

3.1.7.3. Non-Functional Requirements of the DT. Identification of the programmatic, resilience, and maintainability requirements derived during the requirement definition and decomposition process, and each assigned a project-unique identifier.

3.1.7.4. Traceability to Requirements. The list of traces from each DT Component, DT Behavior, and DT Interface identified within this deliverable to the DT requirements allocated to it.

3.1.7.5. Traceability from Requirements. The list of traces from each DT requirement to the DT components, DT Behavior, DT Interface, DT Use Cases, and Digital Threads to which it is allocated.

3.1.8. Digital Twin Verification, Validation, and Accreditation. The description shall include the verification, validation, and accreditation information unique to the DT context. The information provided may be documented in a Test and Evaluation Master Plan (TEMP), but shall contain the following:

3.1.8.1. Verification Plans. Identification of the plans that will be used to verify and validate the implementation and requirements of the DT and each assigned a project-unique identifier. A plan includes the following:

3.1.8.1.1. Type of Verification (e.g., live, predicted, simulated, etc.)

3.1.8.1.2. Quality of Required Data (e.g., level of fidelity, significant figures, or simplifying assumptions)

3.1.8.1.3. Range of conditions necessary to perform verification test

3.1.8.1.4. Criteria for an acceptable result.

3.1.8.1.5. Plan Traceability. The list of traces from each Plan to a User Story, Use Case, or Digital Thread.

3.1.8.1.6. Criteria Traceability. The list of traces from each Criteria to a performance condition stated in a requirement.

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3.1.8.2. Accreditation Information. Identification of the information and information type (e.g., model, data, documentation, etc., and the IP rights necessary to use or access the information) necessary to ensure the consistent accreditation of each DT (and instances thereof) across the lifecycle of the system of interest.

3.1.8.3. DT Measures of Performance. Identification of the measures or attributes necessary to assess and verify the accuracy of DT generated content.

3.1.8.4. Approach to Representing Phenomena. Identification and description of the approach for credibility (or confidence) thresholds that a DT (or instances thereof) must meet while representing the modeled phenomena (physical, behavioral, etc.) of the system of interest under relevant environmental conditions and each provided a project-unique identifier. Including the risk of uncertainty to model performance compared to the represented phenomena and the identification of what DT Measures of Performance are used to make that assessment.

3.1.8.5. Approach to Representing System of Interest. Identification and description of the approach for credibility (or confidence) thresholds that a DT (or instances thereof) must meet while representing the system of interest and each provided a project-unique identifier. Including the risk of uncertainty to DT performance compared to the physical system and the identification of what DT Measures of Performance are used to make that assessment.

3.1.8.6. Approach to Representing Build-Up. Identification and description of the approach for credibility (or confidence) thresholds that a DT (or instance thereof) must meet while representing the build-up (manufacturing, supply chain, etc.) of the system of interest and each provided a project-unique identifier. Including the risk of uncertainty to model performance compared to the represented context and the identification of what DT Measures of Performance are used to make that assessment.

3.1.8.7. Approach to ASOT Confidence. Identification and description of how the DT asserts its confidence as an Authoritative Source of Truth (ASOT) throughout its life, including what test plans support those assertions on a continuing basis.

3.1.8.8. Approach to Documentation Within the Digital Engineering Ecosystem (DEE). Identification and description of how the DEE (such as an Integrated Development Environment for Software Development) documents VV&A evaluations of the DT and descriptions for how it is linked to inform program decisions.

3.1.9. Information Technology Infrastructure. The description shall identify the IT Infrastructure necessary to support the deployment, operation, maintenance, evaluation, and protection of the DT, its interfaces, and, as applicable, interfacing

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components. The infrastructure described will be considered prerequisites for the deployment of the delivered Digital Twin.

3.1.9.1. Components of the Requisite IT Infrastructure. HWCIs, CSCIs, data repositories, and networking for each phase of a represented system's lifecycle and each assigned a project-unique identifier. Include resource requirements and constraints such as power and memory. Include Contract Data Requirements Lists (CDRLs) or DIDs applicable to each IT Component.

3.1.9.2. Initial Storage Requirements for the DT. Both non-recurring and recurring. Non-recurring storage requirements are expected to define the storage necessary to host and operate the DT. Recuring storage requirements are expected to define the storage necessary to retain the data created or collected over the course of DT operations. Including identification of redundancy and backup requirements to ensure availability of the DT.

3.1.9.3. Resource Growth Projections. Identification of the Storage Requirements that have the capacity to change over the life of the DT, the specific metrics that inform that change, and the extent of that change. Historical trends of similar technology development may be used.

3.1.9.4. Static Relationships of the IT Components. Identification, including a diagram of the static (such as "consists of") relationship(s) of the IT components. Multiple relationships may be present.

3.1.9.5. Data Handled by the IT Infrastructure. Identification of the data that will be shared across the IT Infrastructure between components of the DT and each assigned a project-unique identifier.

3.1.9.6. Data Lifecycle. Identification and description, including a diagram, of the lifecycle of each identified data and its flow through the infrastructure. Multiple flows may be present, and the flow may change as the DT matures through the represented system's lifecycle. Include resource requirements and constraints to sustain the end-to-end flow and its scalability.

3.1.9.7. Purpose of Each IT Component and Data Handled. Identification of the DT requirements allocated to it.

3.1.9.8. IT Licenses Needed: Identification of the license(s) and the IP rights that will be needed (at an ongoing basis) to retain access to and use of each component of the IT Infrastructure, as applicable.

3.1.9.9. Lifecycle Needs for each Data Repository Component within the IT Infrastructure. For each data repository component within the IT Infrastructure supporting DT needs, include descriptions for development, deployment, maintenance, and access needs.

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3.1.10. Security and Protections. The description shall describe the risks, threats, and controls unique to the DT context. The information provided may be documented in a Security Plan or System Engineering Management Plan (SEMP), but shall contain the following:

3.1.10.1. Risks and Threats to DT Confidence. Identification of the risks or threats that could compromise the confidence in or operations of the DT throughout its life (e.g., conflicting data, acquired CSCI, faulty HWCI, incomplete model, etc.), particularly in its capacity as an ASOT. Including the likelihood of occurrence over the DT's life and the severity of the impact to DT confidence or operations. Assign each a project-unique identifier.

3.1.10.2. Controls on the DT. Identification of the mechanisms, techniques, technologies, or various solutions implemented to secure and protect the DT against risks and threats throughout its life (called Controls) (e.g., physical controls, cybersecurity, data protections, operational controls, etc.), including its purpose, where it is implemented within the DT or IT Infrastructure context, and each assigned a project-unique identifier.

3.1.10.3. Controls Traceability. The list of traces from each Control to the risks or threats the Control intends to protect against, the confidence to which the Control will sustain its protection over time (or over discrete events), how that protection can be tested and validated over the DT's life, and who will be validating that protection.

3.1.10.4. Risk and Threat Growth Projections. Identification of the risks and threats that have the capacity to change over the life of the DT, the specific metrics that inform that change, and the extent of that change. Historical trends and technology developments may be used. Include when those risks and threats exceed the capabilities of implemented Controls.

3.1.10.5. Software and Hardware Bill of Materials (S/HBOM). Identification of the supply chain and heritage of all DT Components, in particular, the third-party or open-source software or hardware used within or in the development of the DT.

3.1.11. Development Methodology. The description shall include a summarization of the unique methodologies used in developing the DT. The information provided may already be documented in some development plan, but shall contain the following:

3.1.11.1. DT and ASOT Development Methodology. Identification of the approach, practices, and processes used during the development lifecycle of the DT (e.g., scope, define, design, build, certify, etc.) and the ASOTs required of the government.

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3.1.11.2. DT Refinement Methodologies. Identification of the approach, practices, and processes used during the refinement lifecycle (e.g., define, decompose, manage, etc.) of the following:

3.1.11.2.1. DT Requirements

3.1.11.2.2. DT User Stories, Use Cases, and Digital Threads.

3.1.11.2.3. DT User Experience(s)

3.1.11.3. DT Monitoring Methodology. Identification of the approach, practices, and processes used during the monitoring lifecycle (e.g., track, report, etc.) of the following:

3.1.11.3.1. Development of the DT (and auxiliary resources) system and its alignment with the physical system.

3.1.11.3.2. Discrepancies between instances, versions, or branches of the DT.

3.1.11.3.3. Schedule of the DT development, deployment, maintenance, and decommissioning (and its instances).

3.1.11.4. DT Verification & Validation Methodologies. Identification of the approach, practices, and processes used during the verification and validation lifecycle (e.g., verification, validation, testing, etc.) of the following:

3.1.11.4.1. Digital Engineering Ecosystem (DEE) so the government may independently reproduce, test, and evaluate the DEE (such as an Integrated Development Environment for Software Development).

3.1.11.4.2. Metrics to verify, validate, and test DT functionality over time.

3.1.11.5. Assurance Methodologies. Identification of the approach, practices, and processes used to ensure the following:

3.1.11.5.1. Incorporation of standards, guidelines, policies, and regulations into the development process.

3.1.11.5.2. Conformance of the DT to cybersecurity and classified system requirements.

3.1.11.5.3. Configuration Management of the DT, evolution of its components (including data elements, CSCI, HWCI, and models) over time, and any alignment to open systems architecture approaches.

3.1.11.5.4. Management of the DT Program (e.g., cost, schedule, risk, etc.)

3.1.11.5.5. Data storage and backups of exchanged data between DT and system, to include periodicity of backup, redundancy, and frequency of data overwrite.

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3.1.11.5.6. Interactions with proprietary content (e.g., tools and data) are controlled and intentional.

3.1.11.5.7. Evaluation (and optimization) of the DT for modularity, extensibility, and resource use.

3.1.12. Digital Twin Deployment. The description shall include the following:

3.1.12.1. Use Case Deployment Tiers. For each Use Case (provided by the government and defined while satisfying requirements from section 3.1.6: User Stories, Use Cases, and Digital Threads), define the criteria to deploy an initial operational capability with additional criteria and milestones (called Use Case Deployment Tiers) as functionality or technology evolves through the life of the DT; each assigned a project-unique identifier.

3.1.12.2. Levels of Deployment. Considering the DT as the sum of all its Use Cases, define key Levels of Deployment, where each level is a unique configuration of Use Case Deployment Tiers of increasing operational capability, and provide a rationale for why that level is being defined (e.g., all Use Cases have achieved an initial operational capability, a new database is being integrated and will affect Use Cases x-z, etc.). Assign a project-unique identifier for each Level of Deployment.

3.1.12.3. Deployment Capabilities. Identification of the tools, infrastructure, and expertise necessary to deploy the DT (or instances thereof) (called Deployment Capabilities) and state its purpose. Each assigned a project-unique identifier.

3.1.12.4. Deployment Methodology. Identification and description of the approaches, practices, and processes used during the deployment lifecycle (e.g., when, where, and how a DT shifts from a development environment to a production environment).

3.1.12.5. Deployment Schedule. Identification and description of the sequence of activities and milestones with dates (called Deployment Schedule) of each instance (and incremental delivery) of the Digital Twin through its life and potential evolutions. For each instance or increment to be deployed through the life of the DT, describe a Deployment Schedule for each portion of the DT that needs to be separately deployed. The Deployment Schedule shall be defined with relative dates (i.e., time periods) if dates are not possible.

3.1.12.6. Entities Responsible for Deployment. For each Deployment Schedule, identify the entity responsible for performing that portion of deployment.

3.1.12.7. Deployment Limitations. Identification of the limitations on when and where DT may be deployed (called Deployment Limitations) and each assigned a project-unique identifier.

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3.1.12.8. Deployment Uncertainty. For each DT that will be deployed, include a quantifiable estimate of the uncertainty and risk to deployment criteria and each assigned a project-unique identifier.

3.1.12.9. Technology Impacts on Future Deployments. Identification of the impact that technology development will have on future Deployment Tiers, Levels, Schedules, and Limitations using historical trends.

3.1.13. Digital Twin Maintenance. The description shall include the following:

3.1.13.1. Types of Maintenance Activities. Identification and description of the types of maintenance activities that would be performed over the lifecycle of the DT (and its infrastructure), and each assigned a project-unique identifier.

3.1.13.2. Maintenance Capabilities. Identification of the tools, infrastructure, expertise, resources, and dependencies necessary to maintain the DT (or instances thereof) (called Maintenance Capabilities) and state its purpose. Each assigned a project-unique identifier.

3.1.13.3. Maintenance Methodology. Identification and description of the approaches, practices, and processes used during the maintenance lifecycle (e.g., personnel training).

3.1.13.4. Maintenance Schedule. Identification and description of the sequence of activities and milestones with dates (called Maintenance Schedule) of each instance of the DT through its life and potential evolutions. For each instance to be maintained through the life of the DT, describe a Maintenance Schedule for each portion of the DT that needs to be separately maintained. The Maintenance Schedule shall be defined with relative dates (i.e., time periods) if dates are not possible.

3.1.13.5. Entities Responsible for Maintenance. For each Maintenance Schedule, identify the entity responsible for performing that portion of maintenance.

3.1.13.6. Impacts from Operational and Maintenance Need Changes. Identification of each change to the operation and maintenance needs of the DT with a project-unique identifier and a listing of the impacts to DT Use Cases, DT Components, and DT Behavior. Including impacts to data quality, data flows, and data structure.

3.1.13.7. Maintenance Limitations. Identification of the limitations on when and how the DT may be maintained and each assigned a project-unique identifier.

3.1.13.8. Maintenance Uncertainty. For each DT that is to be maintained, include a quantifiable estimate of the uncertainty and risk to maintenance criteria and each assigned a project-unique identifier.

3.1.14. Digital Twin Decommissioning. The description shall include the following:

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3.1.14.1. Decommissioning Criteria. Identification and description of the conditions when instances of or every instance of a DT will be considered for decommission and each assigned a project-unique identifier.

3.1.14.2. Decommissioning Plans. Identification of the plans that will be used to decommission the implementation of the DT and each assigned a project-unique identifier.

3.1.14.3. Decommissioning States. Identification of the various states of decommissioning the DT can be in and each assigned a project-unique identifier.

3.1.14.4. Archival Standards of the DT. For each state of decommissioning, list (as applicable) the archival standards the state conforms to, the minimum and maximum time a DT shall stay in that state, the conditions that shall transition a DT between states, the conditions that shall transition a DT back to full operational capability, and accessibility of features and data.

3.1.15. Interoperability. The description shall include the following:

3.1.15.1. Methodologies for Interoperability Assurance. Identification of the approach, practices, and processes that ensure metadata, data, CSCI, and HWCI interoperability and each assigned a project-unique identifier.

3.1.15.2. Instances of Interoperability. Identification of interoperability between uniquely deployed DT of the system of interest (or parts thereof), uniquely deployed DT of other systems, (or parts thereof), the system of interest, and other virtual or physical assets and each assigned a project-unique identifier.

3.1.15.3. Characteristics of the Interoperability. For each instance of interoperability, describe what is needed to enable the interoperation, the limitations of the interoperability, and the protections that support the protection of constituent components.

3.1.15.4. A list of all incoming data from the system-of-interest to the DT, the system component that created the data, and what information (if any) was used to create that data. Each assigned a project-unique identifier.

3.1.15.5. Programmatic Traceability of Incoming Data. A list of how the incoming data to (or usage of) the DT is in accordance with the identified use cases, cybersecurity protections, and model development plan.

3.1.15.6. A list of all the outgoing data to the system-of-interest from the DT, the DT component that created the data, and what information (if any) was used to create that data. Each assigned a project-unique identifier.

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3.1.15.7. Programmatic Traceability of Outgoing Data. A list of how the outgoing data from (or usage of) the DT is in accordance with the identified use cases, cybersecurity protections, and model development plan.

3.1.15.8. Resources Involved During Interoperability. Identification of the information, data, or power used or processed through the DT, and each assigned a project-unique identifier.

3.1.15.9. Lifecycle for Each Resource. Including a description and diagram of the lifecycle(s) of that resource. Multiple lifecycles may be represented as the DT matures.

3.1.15.10. Interfaces Necessary for Interoperability. Descriptions of all system-of-interest interface definitions and any unique interfaces developed to properly operate and maintain the integrated models and systems, including receiving updates to system parameters (system component status and configuration) when appropriate.

3.1.16. User Experience. The description shall include the user experience information unique to the DT context. The information provided may be documented in a dedicated Instruction or Manual, but shall contain the following:

3.1.16.1. User Accessibility. Identification of the standards that ensure the DT and auxiliary resources (e.g., User Manual, User Interface(s), and User Training) are accessible to identified Users.

3.1.16.2. User Manual. Includes detailed descriptions of the computer operational environment, connectivity needed to external environments (including connections to external ASOTs for use cases where up-to-date system status and configuration will be needed), and any third-party commercial software and tools (with any needed plug-ins, etc.) needed to operate the DT.

3.1.16.3. User Interface(s). Descriptions of the User Interface(s) used in the development, deployment, operation, and maintenance of the DT including dashboards and visualizations to interrogate, visualize, and interact with the DT and associated data.

3.1.16.4. User Training. Descriptions of the learning materials, instructor lead classes, or practical simulations on how to use DT.

3.2. Implementation Content. This content describes the select types of requirements that shall be imposed on a specific DT to enable the functional capability assigned to that DT. To the extent that the following functions are required for the DT, this shall include:

3.2.1. System Component Status. How the DT incorporates the known operational and maintenance status of applicable system components throughout their life cycles,

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when authorized and connected to authoritative sources of such operational and maintenance status.

3.2.2. Analysis Support - Interface & Integration Trades. How the DT provides integration and interface analysis/test with systems and system components outside of the program on contract.

3.2.3. Analysis Support – Operator Decision Trades. How the DT provides system operations trades and operator decision support.

3.2.4. Analysis Support - Sustainment, Maintenance, and Logistics Trades. How the DT provides sustainment trades, sustainment design, maintenance decision support (including predictive maintenance), and logistics analysis.

3.2.5. Analysis Support - Enterprise Trades. How the DT provides enterprise operations trades and decision support conducted by external organizations that use models for real-time operations. Note: this shall include system modeling supporting activities.

3.2.6. Analysis Support - As-Is Model for the Future Enterprise. How the DT provides "as is" models supporting future enterprise architecture and enterprise operational planning and analysis that use or potentially use the System of Interest (SOI).

3.2.7. Analysis Support – Could-Be Systems. How the DT provides a foundation for future system iterations, versions, or evolutions.

3.2.8. Deployment Threshold. How the DT meets deployment thresholds over the System of Interest's designated design life, by deployer personnel having the Deployment Capabilities indicated in Section 3.1.12.3.

3.2.9. Deployer Training. How the DT provides initial and sustained deployer training.

3.2.10. Operation Threshold. How the DT meets operation threshold over the System of Interest's designated design life.

3.2.11. Operator Training. How the DT provides initial and sustained training to operator personnel.

3.2.12. Maintenance Threshold. How the DT meets maintenance thresholds over the System of Interest's designated design life, by deployers having the Maintenance Capabilities indicated in Section 3.1.13.2.

3.2.13. Maintainer Training. How the DT provides initial and sustained training to maintenance personnel.

3.2.14. Model Credibility - Initial Accreditation. How (and to what fidelity) the DT represents the System of Interest at the moment of System delivery.

3.2.15. Model Credibility - Fitness. How (and to what fidelity) the DT represents the System of Interest at regular intervals and over a specified (discrete or continuous) range of stakeholder events.

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3.2.16. Model Credibility - External. How (and to what fidelity) the DT represents the System of Interest at regular intervals and over a specified (discrete or continuous) range of external environmental interactions.

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